

Lecture

Pattern Matching and Machine Learning for

Audio Signal Processing

MA-INF 2212

§1 Introduction and Motivation

Prof. Dr. Frank Kurth



...typically deals with the tasks of

- representing,
- analyzing / obtaining information from,
- processing / transforming, or
- generating / synthesizing

acoustic data on a computer.

In our research (@ U Bonn & Fraunhofer FKIE), we are mainly interested in the first three topics.

Roughly, we are interested in

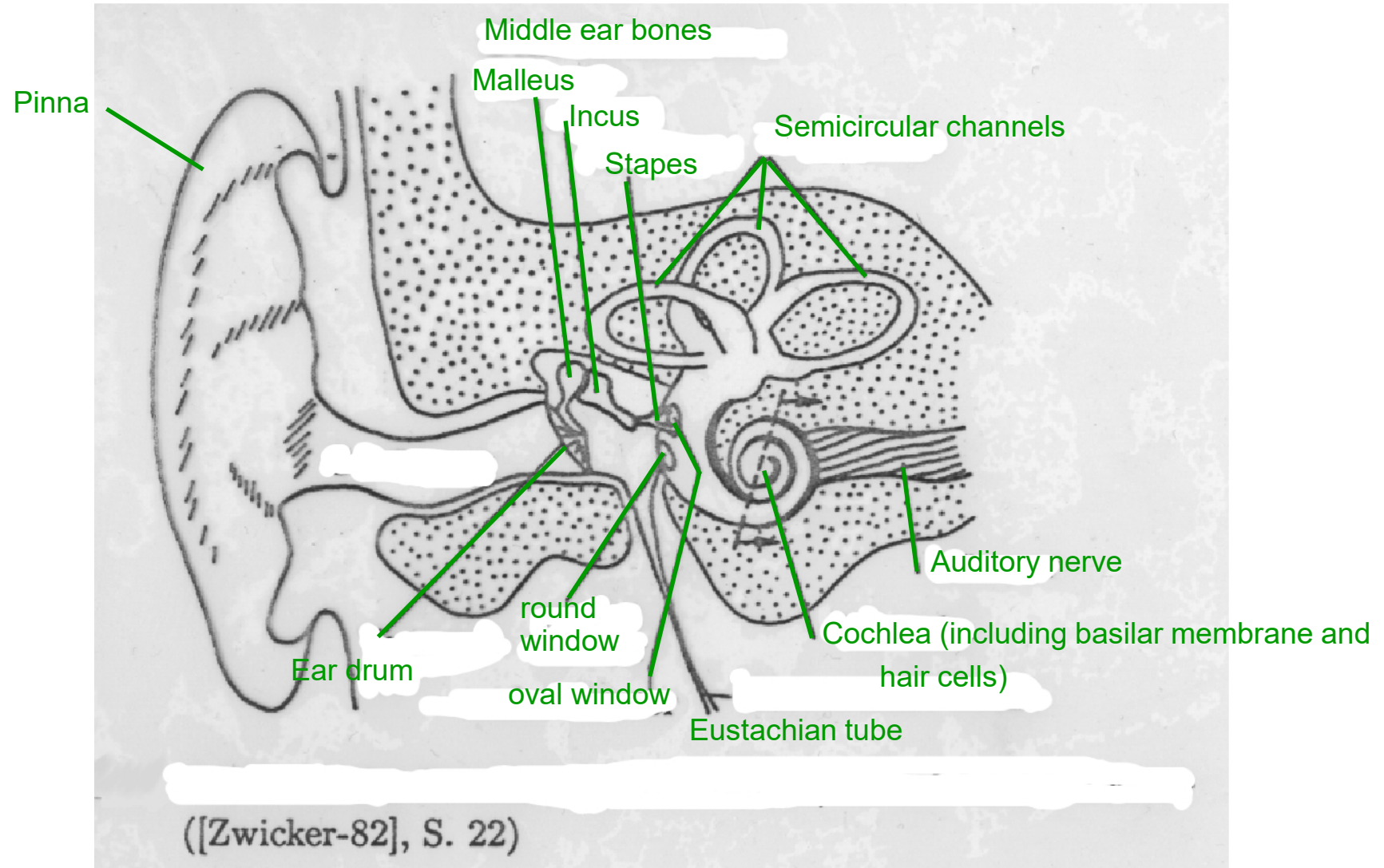
„information a human can perceive with his auditory system“,

the *Human Auditory System* (for short: HAS).

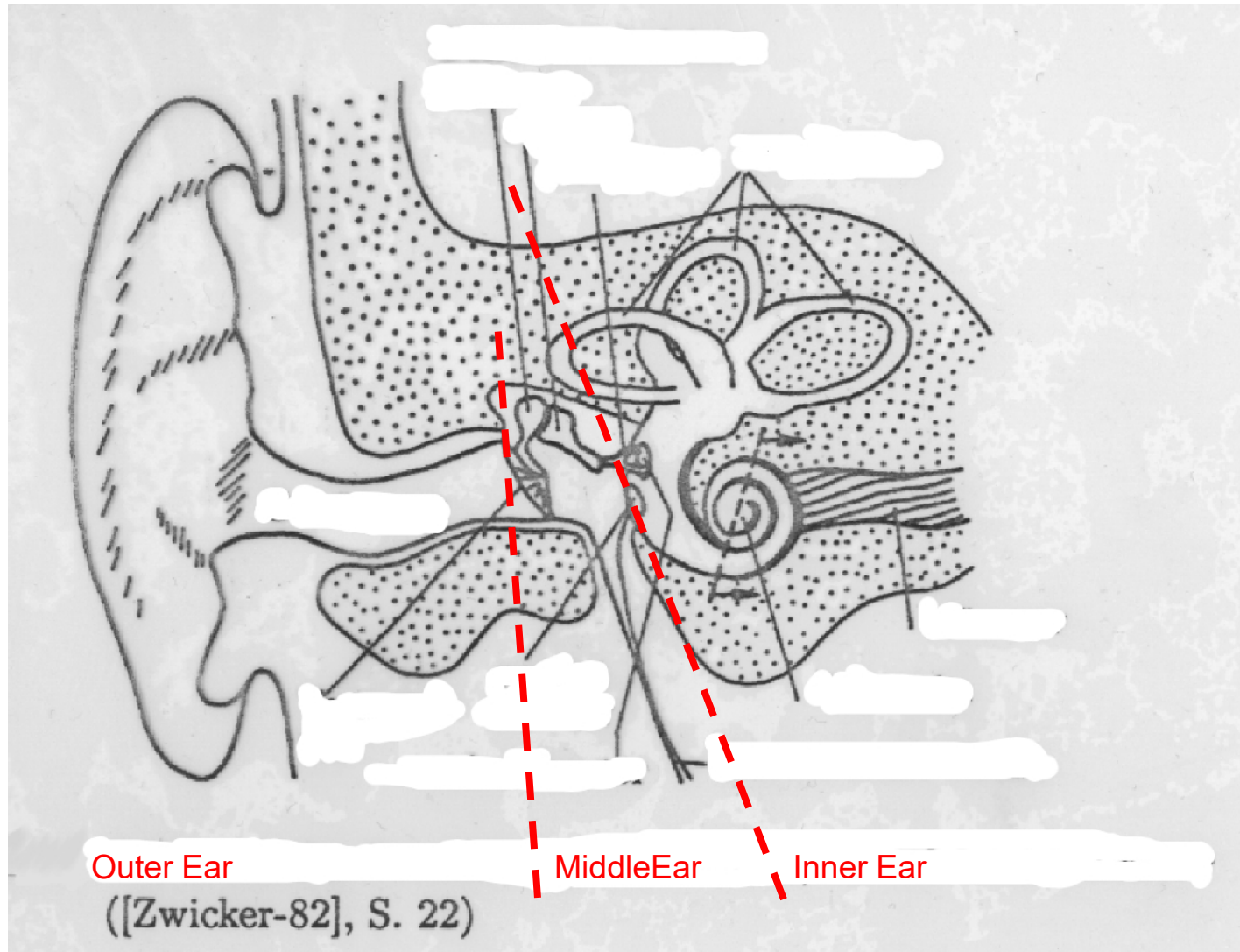
We simply speak of *audio information* or *audio data*.

Do you know some examples?

The Human Auditory System



The Human Auditory System

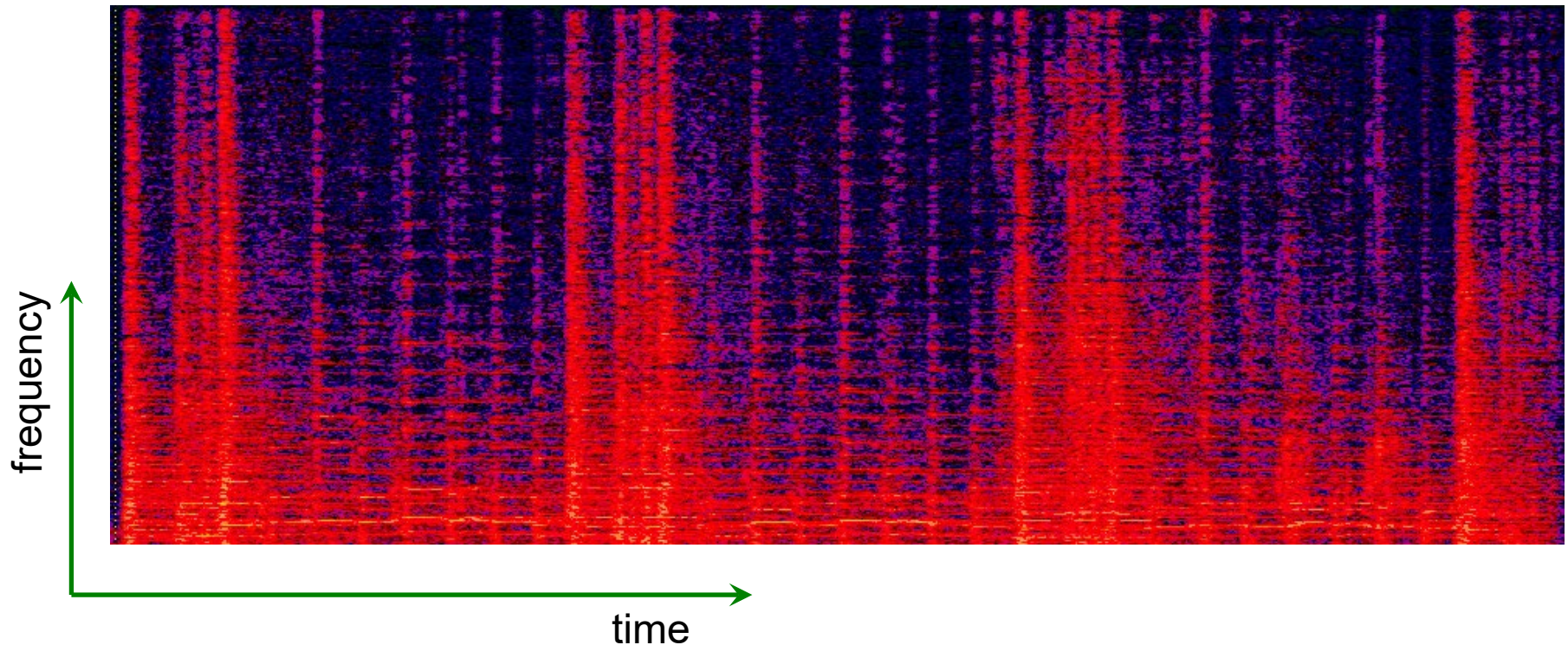


Let's listen to some audio signals

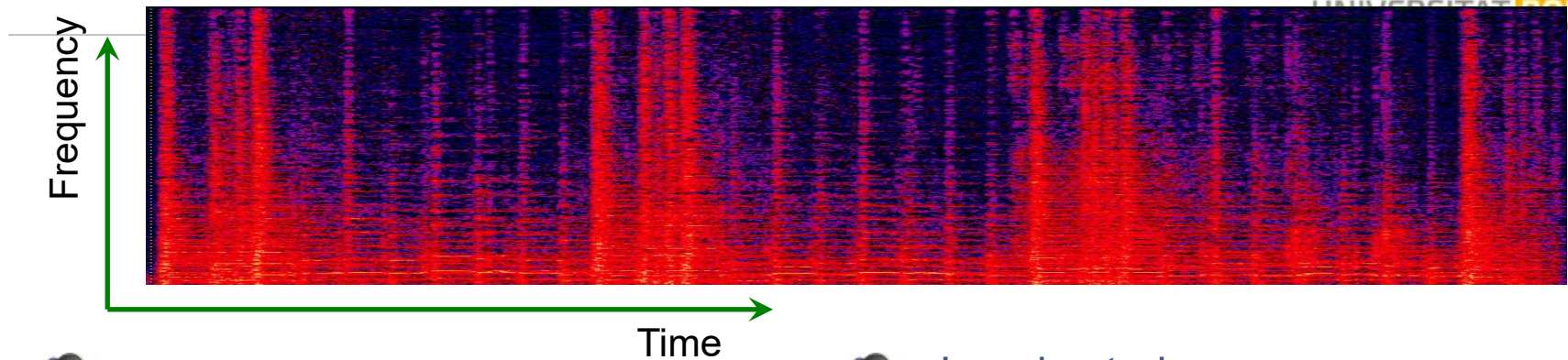
- Speech
- Music
- Animal sounds
- Environmental sounds
- Just noise (what is noise, anyway?)
- ...



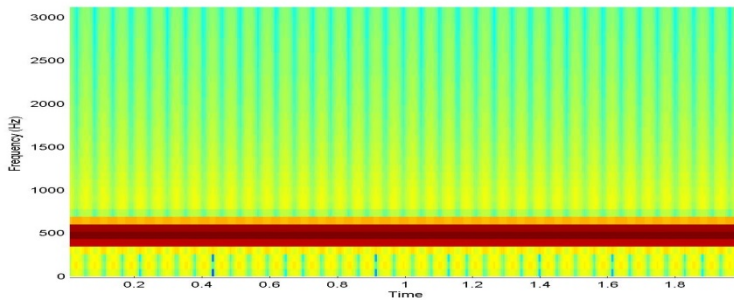
Quiz: Can you **see** the audio example?



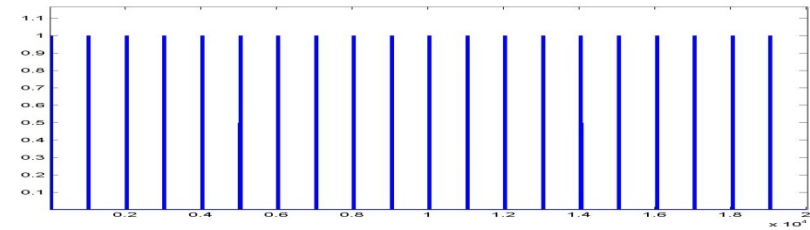
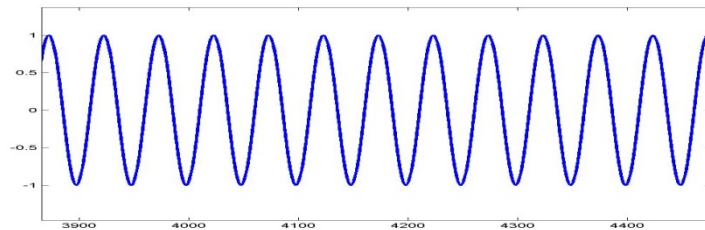
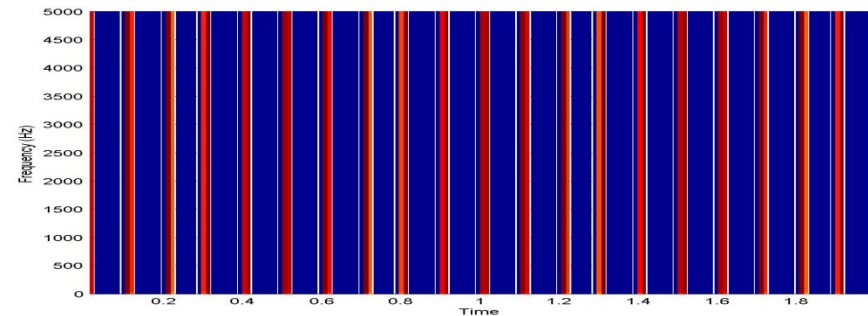
Spectrogram / STFT (§1)



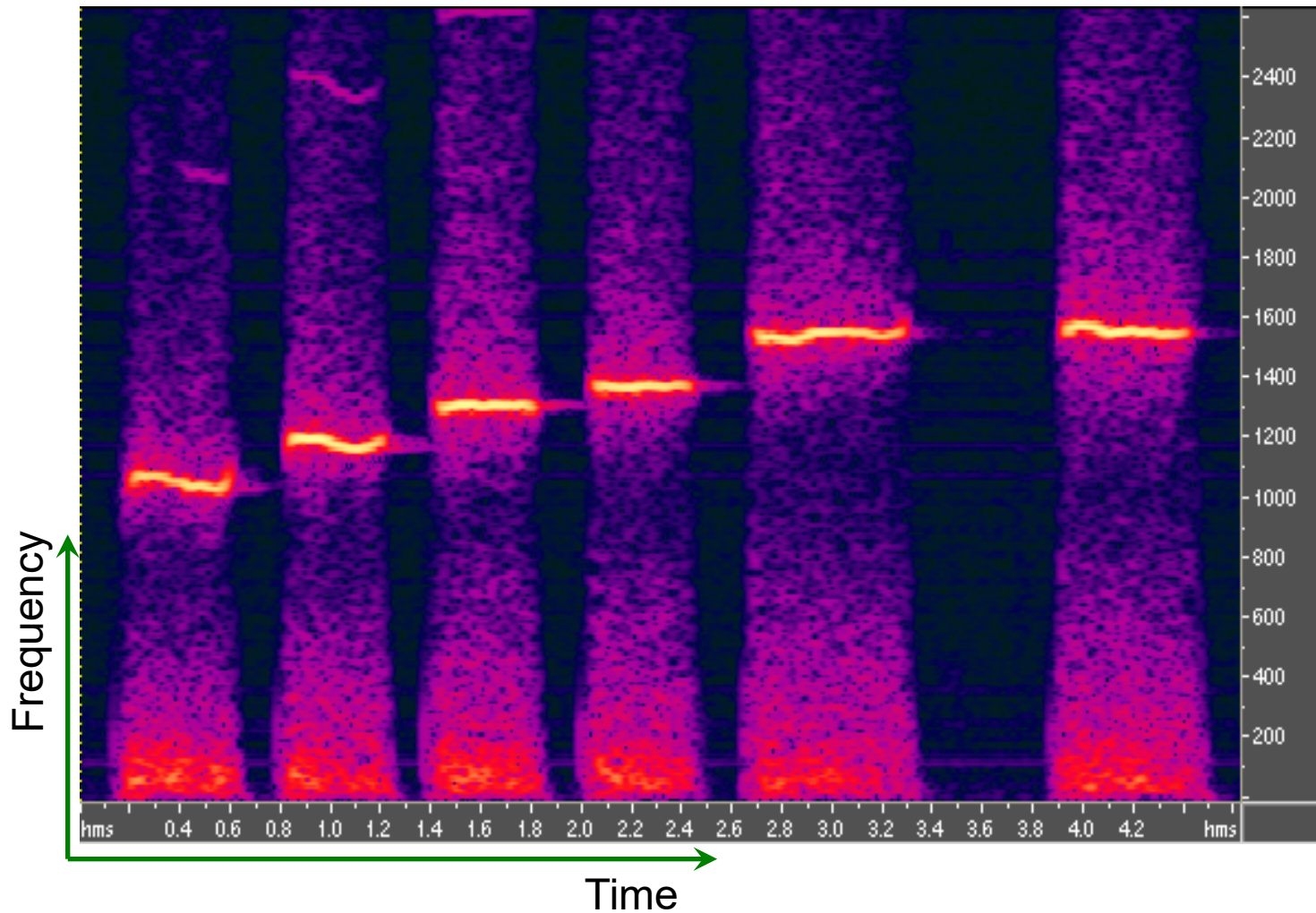
Sinusoid, e.g: 440 Hz



Impulse train



„A melody“ – Sequence of sinusoids



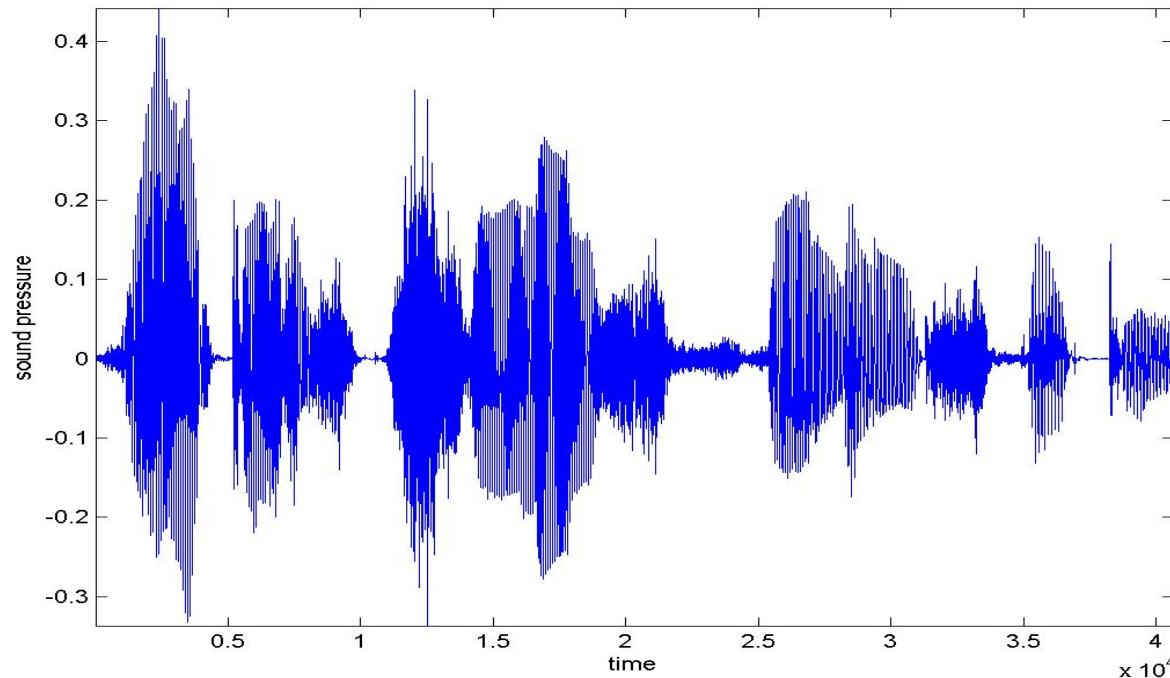
What kind of representations of audio data do you know?

- Representations of the physical waveform
 - Audio CD or DVD
 - Analog tape
 - Record (LP, phonograph/gramophone/vinyle record)
 - .wav – file on a computer
 - compressed format like .mpeg, .mp3, .aac

Those are signal-based representations.

- Meta-formats which can be used to create audio:
 - Texts
 - Sheet music
 - MIDI / MusicXML

Audio Signals



Physical waveform of the audio – sound pressure level over time.

Mathematically, a signal is a function

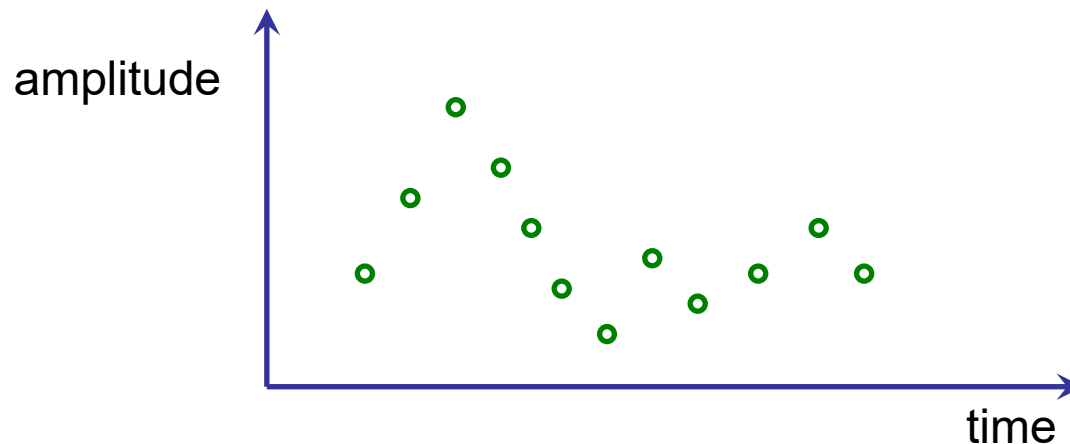
$$f : R \rightarrow R$$

mapping time- to sound pressure-values. ($R \sim$ real numbers)

On a Computer: Discrete Signals

To store audio signals on a computer, the domain and range have to be discretized. A signal now is an integer-valued function

$$x: \mathbb{Z} \rightarrow \mathbb{Z}$$



A signal is mostly notated as a *sequence* of integer values, e.g.

..., -1, -2, -5, -3, 1, 2, 4, 7, 2, -1, ...

could be contents of a .wav-File.

(Here: 1-channel- or monophonic (*mono*-) signal.)

- How to...
 - formally represent audio (as a signal)
 - process/manipulate/transform signals
- Most work in Audio – at least as long as processing of the „physical“ representations is involved – is based on *Signal Processing*.
- On a computer, *Digital Signal Processing*.
- A major part of this lecture will hence be devoted to an introduction to digital signal processing.
- But let's first consider some „real-world“ tasks in audio processing!

Fundamental areas: Pattern Matching and Machine Learning

- Pattern Matching (PM)

„Find known patterns in large, noisy data sets.“

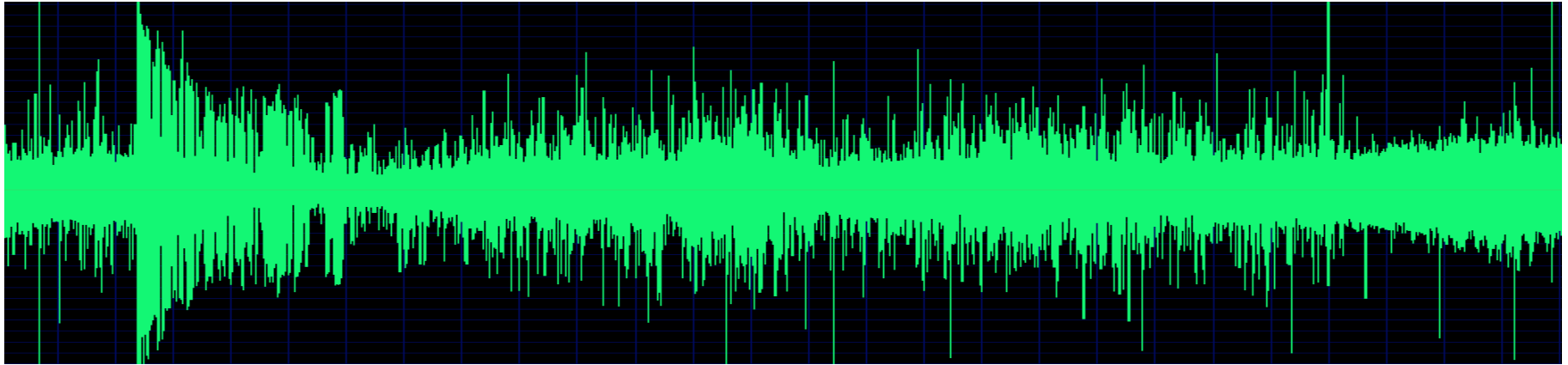
- Detection Tasks
- Precise Localization of patterns or multiple occurrences
- Multiple Patterns can be used to solve classification tasks

- Machine Learning (ML) - a glance at Wikipedia, 11.04.2018

“Machine learning is a field of computer science that uses statistical techniques to give computer systems the ability to “learn” (i.e., progressively improve performance on a specific task) with data, without being explicitly programmed.”

Attributed to

Samuel, Arthur (1959). "Some Studies in Machine Learning Using the Game of Checkers". IBM Journal of Research and Development. 3 (3).



Given: An audio signal and a target signal type

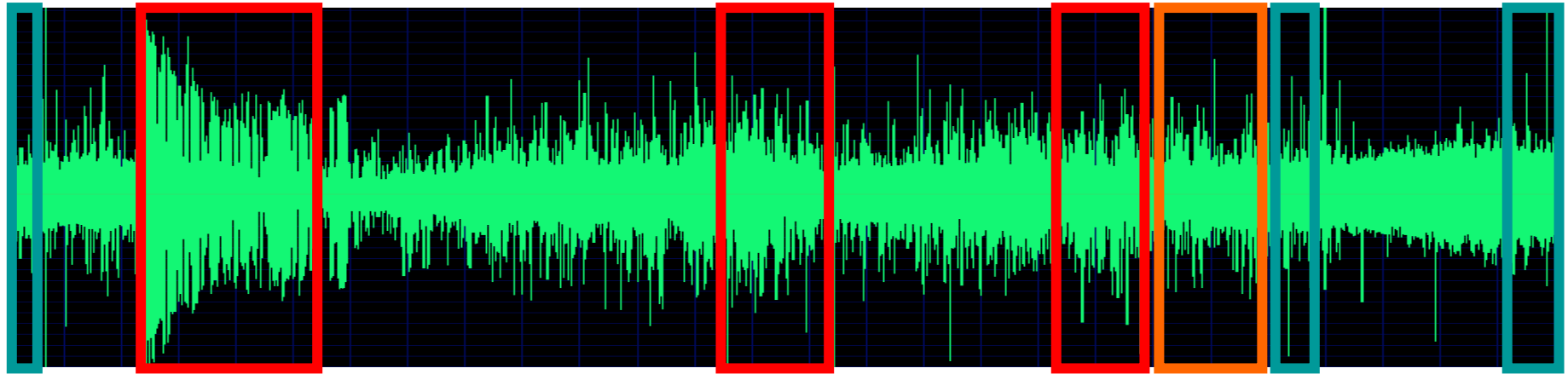
Task: Return a list of (temporal) signal segments containing a signal of the targeted type

Side conditions:

- Return all such segments (no „false negatives“).
- Do not return wrong segments (no „false positives“).

Motivating examples: Audio processing tasks

(1) Audio detection



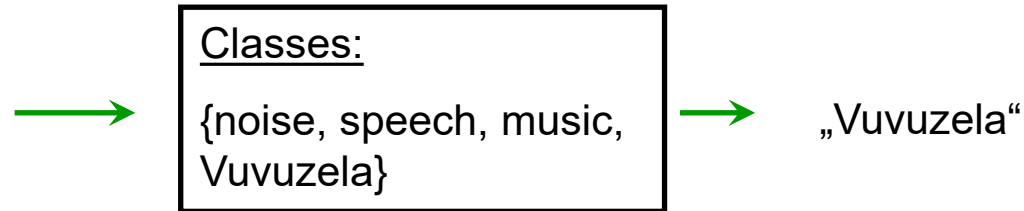
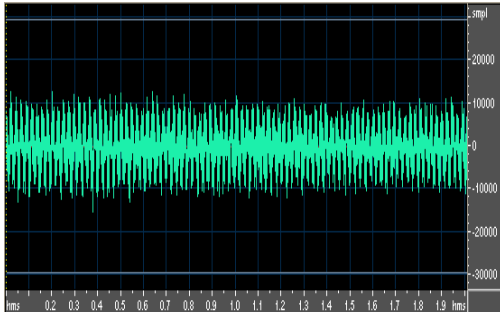
„Speech“ Segments

Also „Speech“ Segments, but weak quality

For comparison: background noise

Motivating examples: Audio processing tasks

(2) Audio classification



Given: Audio signal x ; N different signal types (called „classes“)

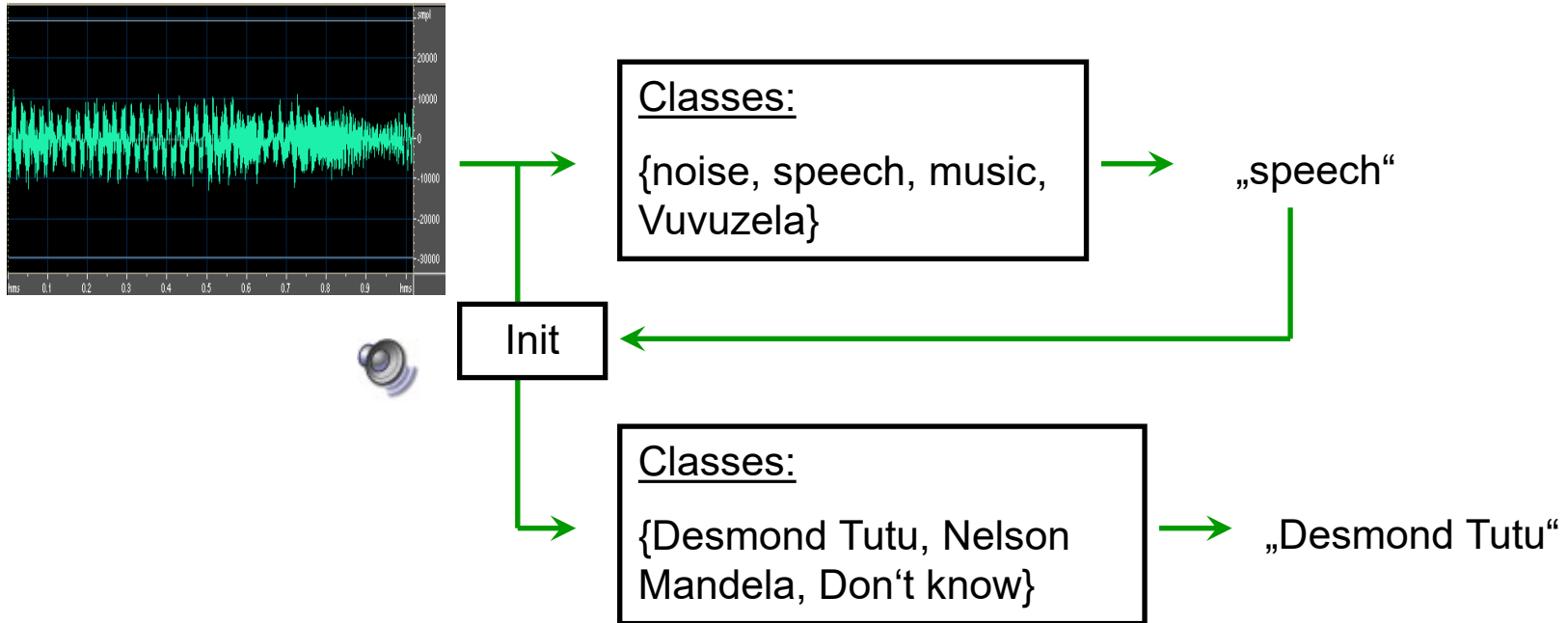
(Class examples: „Peters voice“, „Carols voice“, „background noise“, „don't know“)

Task: Assign to x the correct class.

Side conditions: Typically, x will be of short duration.

Motivating examples: Audio processing tasks

(2) Audio classification – 2-stage classifier



Given: Audio signal x ; N different signal types (called „classes“)

(Class examples: „Peters voice“, „Carols voice“, „background noise“, „don't know“)

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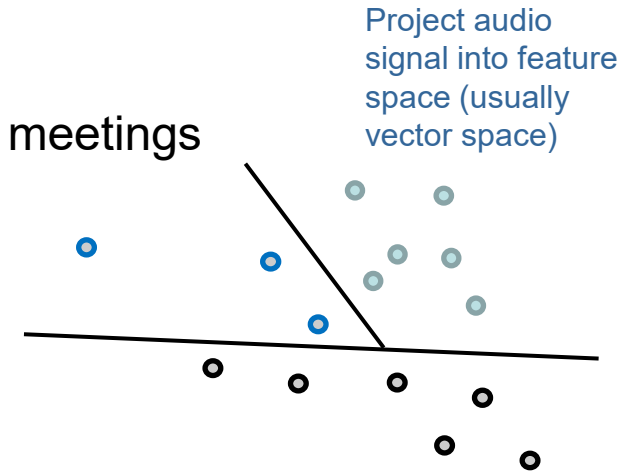
Side conditions: Typically, x will be of short duration.

Motivating examples: Audio processing tasks

(2) Audio classification – further examples

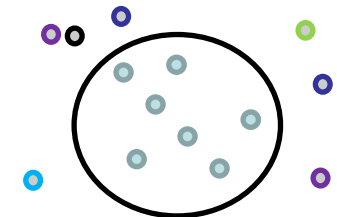
■ Closed-set Speaker classification

- „Who speaks when?“, e.g. for diarization of meetings
- Robustness w.r.t.
 - room acoustics,
 - confusion of speakers  within set,
 - indoor-noises,...



■ Forensic speaker verification

- „Confirm speaker's ID with high confidence.“
- Robustness w.r.t.
 - confusion of  with *any other* person
 - possibly severe noise



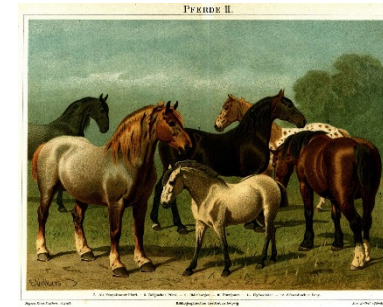
Note: Importance and challenge of open-set problems!

Motivating examples: Signal processing tasks

(2) Classification – Importance of problem specification

■ Classification of horses and zebras in images

- Distinguish any zebra from any horse
- Robustness?



Images showing Zorses (cross between Zebra and Horse)

[Von Fährtenleser - Eigenes Werk, CC BY-SA 4.0,

<https://commons.wikimedia.org/w/index.php?curid=83478158>]

[Von Kumana @ Wild Equines - originally posted to Flickr as carly, CC BY 2.0,

<https://commons.wikimedia.org/w/index.php?curid=5175516>]

[Von Olliver in der Wikipedia auf Englisch - Übertragen aus en.wikipedia nach Commons., Gemeinfrei, <https://commons.wikimedia.org/w/index.php?curid=1974602>]

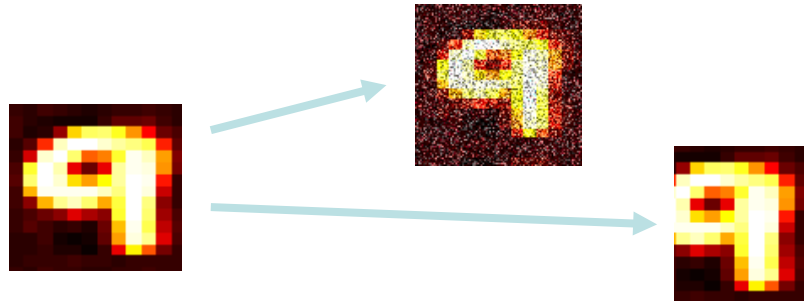
Motivating examples: Signal processing tasks

(2) Classification – Typical Requirements

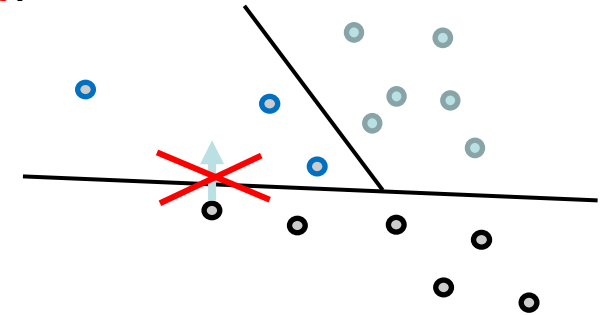
- To what kinds of changes could a classifier be **robust**?

- Two major classes

- Additive noise and corruption

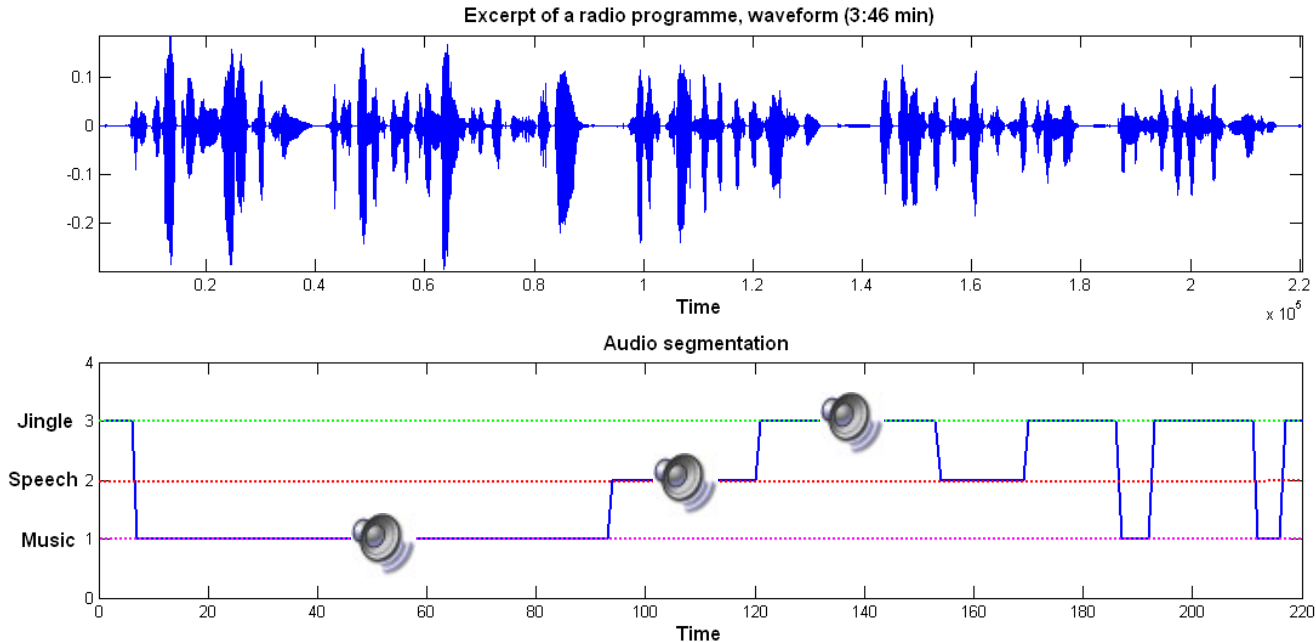


- „Semantically“ meaningful variations



Motivating examples: Audio processing tasks

(3) Audio segmentation



Given: A (typically long) audio signal $x = (x_1, \dots, x_n)$.

Task: Give a (temporal) partition of x in k segments $s_0, \dots, s_{k-1}$ where

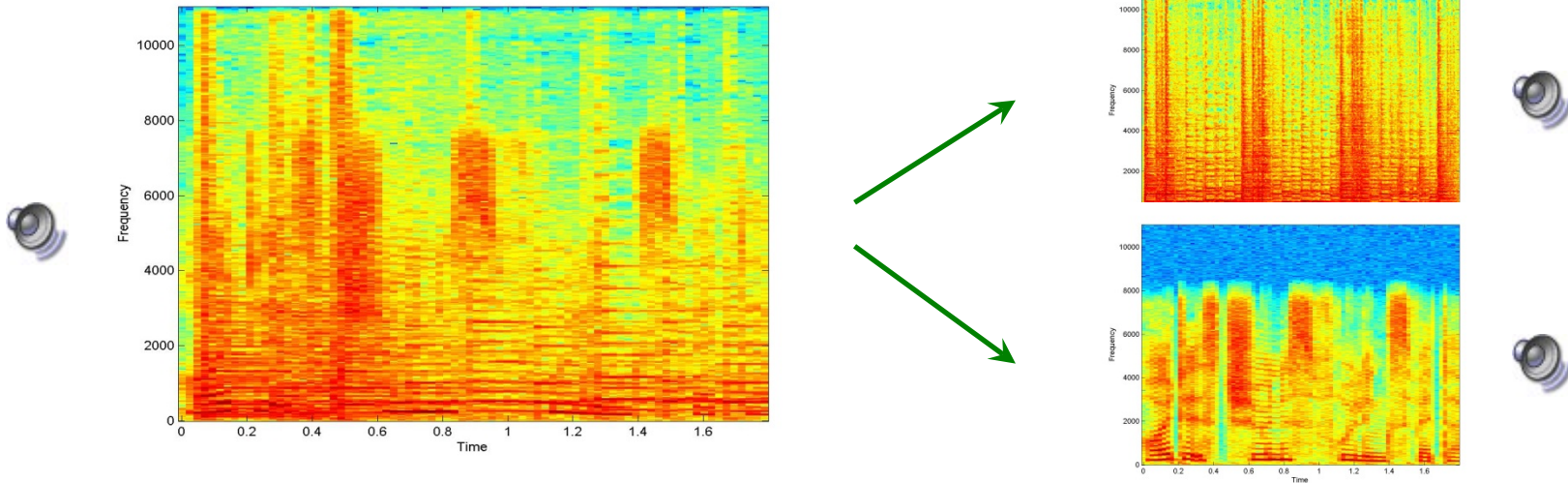
$$s_i = (p_i, p_{i+1}-1) \text{ for integers } 1 = p_1 < p_2 < \dots < p_k = n$$

Side conditions: For example,

- each segment should represent one of N given classes and
- pairs of subsequent segments should belong to two different classes.

Motivating examples: Audio processing tasks

(4) Source separation



Given: N simultaneous audio recordings of a mix of M audio sources

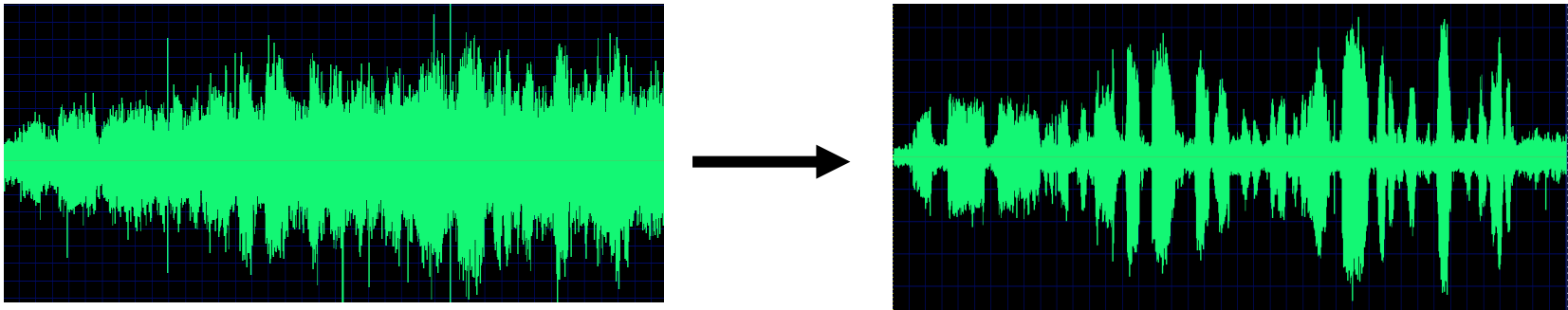
Task: Return all M sources in M separate signals, each containing only one of the M sources.

Remarks:

- Typically, $N < M$.
- Example $N = 1$ is very hard 
- Is the problem well-posed? Assume $15 = a + b$ - what are a , b ?
- ... as sources are generally temporally correlated, there is some hope that the separation problem can be solved for particular cases.

Motivating examples: Audio processing tasks

(5) Noise reduction



Given: A target signal corrupted by noise.

Task:

- Reduce the amount of noise or
- Extract the target signal or
- Increase the intelligibility of the target signal.

Side conditions: Do not reduce the „quality“ of the target signal.

Motivating examples: Audio processing tasks

(6) Audio compression

Given: A target signal x , represented using N bytes.

Task: Convert the signal to a compressed form $c(x)$ requiring $M \ll N$ bytes.

Side conditions: It should be possible to ...

- (perfectly) reconstruct x from $c(x)$ [lossless compression]

$x = (1, 1, 1, 1, 1, 1, 2, 2, 2, 3, 3, 5, 5, 5, 5, 3, 3, 3, 2, 2, 1, 1, 1, 1)$

$c(x) = ((1,6), (2,3), (3,2), (5,4), (3,3), (2,2), (1,4))$

removes **redundance**,

or

- perceptually reconstruct x from $c(x)$ [lossy compression]

x given as a .wav – file „16 bit, 44100 Hz“ (e.g. 544 kB)

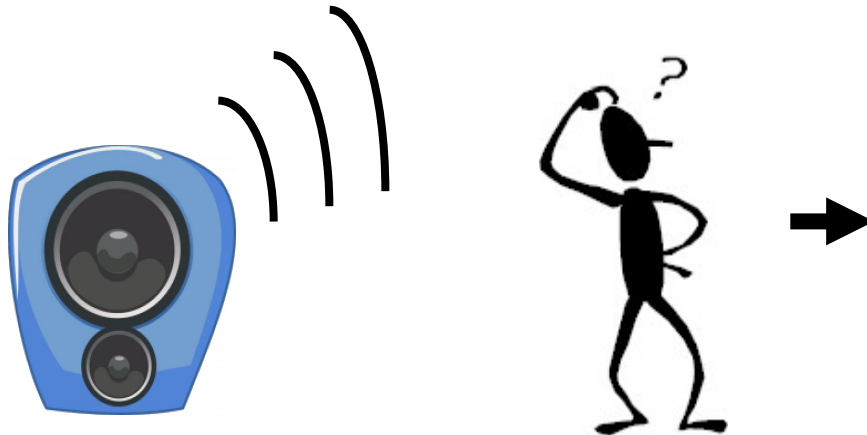
$c(x)$ obtained as a .mp3 – file „@128 kbps“ (e.g. 102 kB)

removes **irrelevance**.



Motivating examples: Audio processing tasks

(7) Audio identification



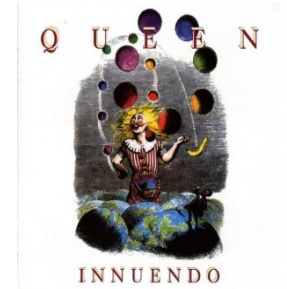
Title: Innuendo

Artist: Queen

Album: Innuendo

Out: 4. Februar 1991

Label: Parlophone (EMI)



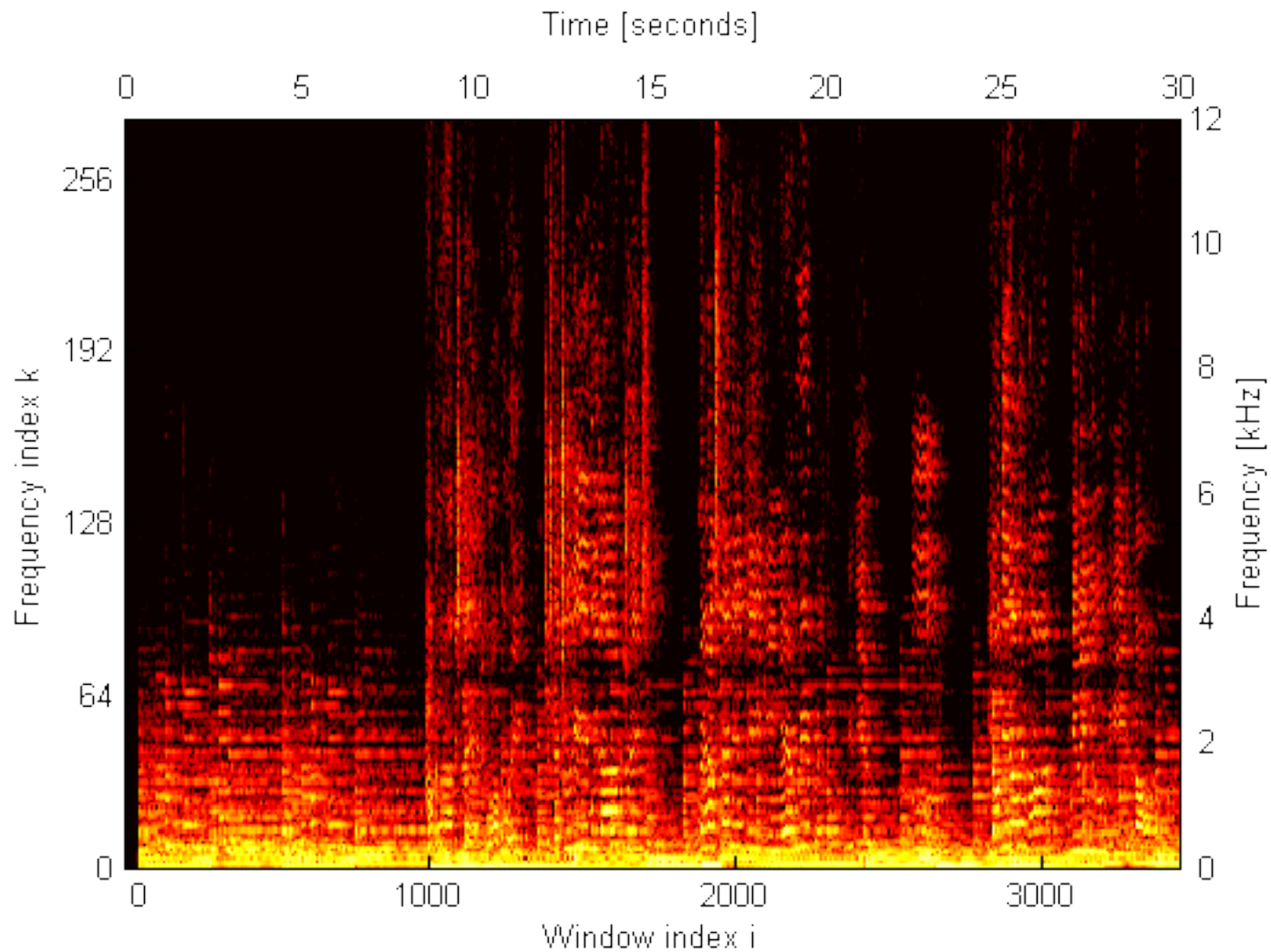
Given: A short music signal, e.g. recorded in a car using a mobile phone.

Task: Find out the name of the piece of music, the interpreter, and the name of the CD.

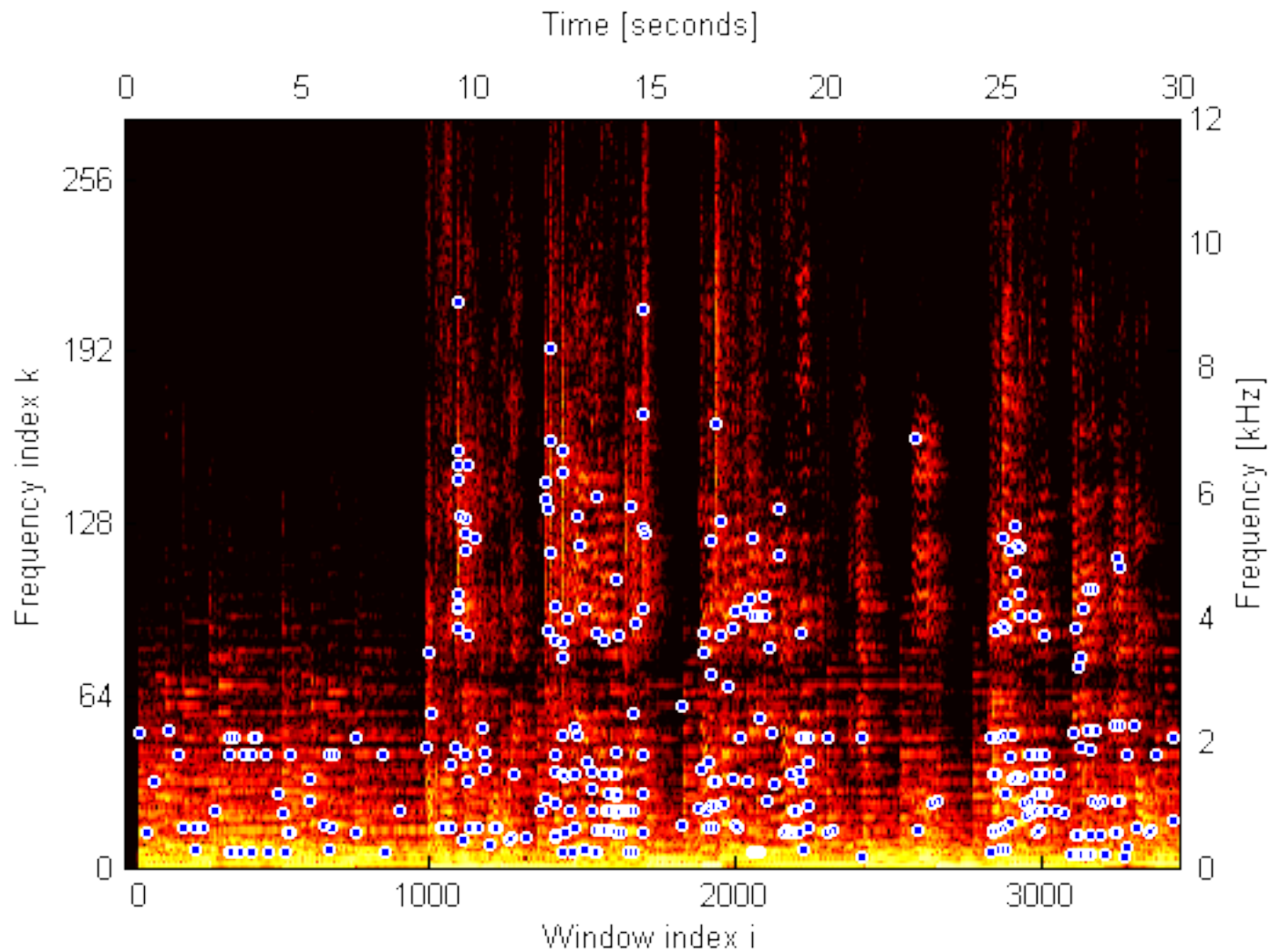
Side conditions:

- This service should be available as a mobile phone application.
- The task should be solved very fast.
- It should work on a very short recording and in a very noisy environment.

Music representation as a spectrogram

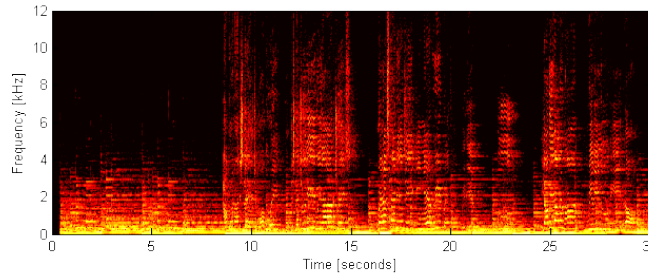


Computation of a digital fingerprint

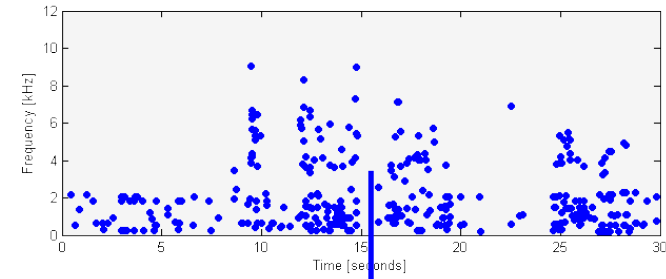


Audio identification ~ data base search

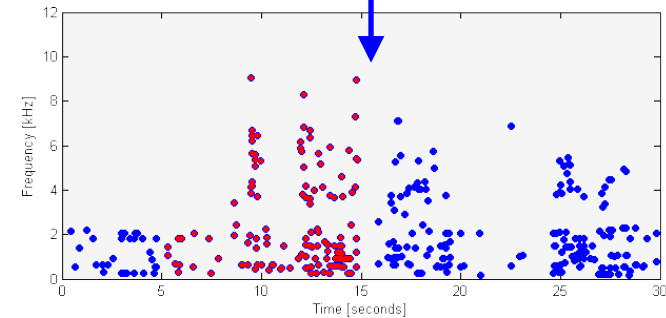
Document



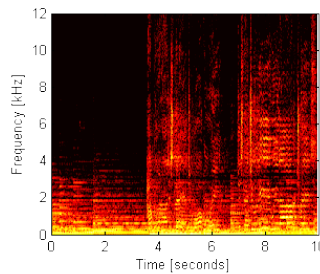
Feature
extraction
→
Indexing



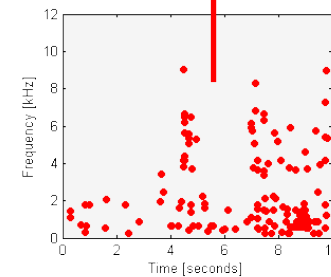
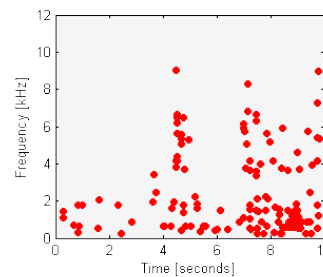
Comparison of
feature sequences



Query



Feature
extraction
→



Match of time-shifted
query sequence

Motivating examples: Audio processing tasks

(7) Audio identification

[2001] AudioID Fraunhofer IIS
Philips Audio Hashing
Audentify Uni Bonn

[2002] Launch of Shazam service

[2009] Shazam as an app

[2011] Top Apps (Apple)
SoundHound (top @paid)
Shazam (4 th @free)

[2012] Titles available
10 Mio. (Shazam)
28 Mio. (Gracenote)



1. Introduction and Motivation
2. Fourier Transforms: DFT, FFT, and STFT
3. Digital Filters
4. Audio Identification
5. Audio Matching
6. Hidden Markov Models (HMMs)
7. Support Vector Machines (SVMs)
8. Harmonic-Percussive Separation (optional)
9. Neural Networks and Deep Learning